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ABSTRACT

The long half-life and the high decay energy of beta sources play a critical role in improving the performance of betavoltaic micro-nuclear batteries. In this study, the pure beta source ^{14}C in the form of powder, which can be transformed into an ultra-thin film by using $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$, was selected to design planar 4H-SiC p-n junction betavoltaic batteries. A comprehensive model was developed utilizing the Monte Carlo code and the COMSOL Multiphysics code to predict the output performance. As a result, based on a 100 μm -thick $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source with a maximum power density of $1.86 \mu\text{W}/\text{cm}^2$ and the current fabrication technology of 4H-SiC p-n junction, we optimized the thicknesses of the p^- -type region ($4.2 \mu\text{m}$) with a doping concentration of $N_a = 3 \times 10^{16} \text{ cm}^{-3}$ and the n^- -type region ($5.8 \mu\text{m}$) with a doping concentration of $N_d = 2 \times 10^{14} \text{ cm}^{-3}$. The corresponding predicted performance values included the short-circuit current density of $0.1 \mu\text{A}/\text{cm}^2$, the open-circuit voltage of 2.15 V, and the maximum power density of $0.2 \mu\text{W}/\text{cm}^2$. Moreover, the energy conversion efficiency of the semiconductor converter can reach 10.6%, while the overall battery efficiency was determined to be 2.9%. Therefore, this research provides a feasible structure for a planar 4H-SiC p-n junction energy converter utilizing the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source and presents a powerful model for predicting the performance of planar betavoltaic batteries.

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I. INTRODUCTION

Betavoltaic nuclear batteries are devices that convert the decay energy of beta sources directly into electrical energy,^{1,2} making them a promising choice for low-power integrated micro-/nanoelectromechanical systems (MEMS/NEMS) in medical implants, sensors in space and deep sea exploration, and special military equipment due to long working lifetime, high power density, large specific energy, and high stability under extreme conditions, such as low temperature.^{3–6} Despite extensive research on enhancing the performance of betavoltaic batteries through theoretical and experimental studies in recent years, there are still several

challenges to overcome in achieving high power density and high energy conversion efficiency.⁵ Currently, increasing attention is being paid to identifying the appropriate sources with high decay energy as they determine the theoretical limit of maximum energy density for betavoltaic batteries.^{4–6}

Compared with typical beta sources, such as ^{147}Pm , ^3H , and ^{63}Ni , the ^{14}C source can show great potential as a high decay energy supply.^{6–8} The ^{14}C source is a pure beta-emitting isotope in the form of powder with an average energy of 49.5 keV and the maximum energy of 156.5 keV.⁸ It has a half-life of 5730 years and can be transformed into diverse compound films.^{7–9} Considering the

radiation damage threshold of semiconductor materials, the energy emitted by ^{14}C source is lower than that required to cause significant damage in most wide-bandgap semiconductors.¹⁰ To date, only limited research on the micro-nuclear batteries utilizing ^{14}C sources has been reported. In one study conducted in 2017 by Alina Akimchenko *et al.*, they designed a betavoltaic energy converter based on SiC through self-organizing mono-3C-SiC endotaxy on a Si substrate. This work highlighted that the irradiated electron-hole pairs with high intensity due to the high decay energy of ^{14}C source in the depletion region can increase the energy conversion efficiency.¹¹ In 2020, a dye-sensitized betavoltaic battery was developed for the first time, utilizing the ^{14}C radioisotope as both the beta source and the counter electrode.⁹ The battery employing the N719 dye demonstrated a maximum power density of 0.095 nW/cm^2 and an energy conversion efficiency of 0.48%. In 2021, Zhao and co-workers conducted simulations to investigate the key factors contributing to energy losses in planar betavoltaic batteries incorporating typical beta sources, such as $^{147}\text{Pm}_2\text{O}_3$, Ti^3H_2 , ^{63}Ni , and ^{14}C in the form of $^{14}\text{C}_5\text{O}_2\text{H}_8$.⁷ Based on their simulated results, it was found that utilizing the ^{14}C source along with wide-bandgap converters is conducive to fabricating betavoltaic batteries with high output performance. However, there has been no reported systematically investigation into aspects such as the self-absorption process of the source and/or the optimized planar semiconductor energy converters, specifically for ^{14}C -based betavoltaic batteries.

Regarding energy converters, it is widely acknowledged that a higher bandgap and advancements in fabrication technology lead to improved output performance.^{2–5} Among the previously developed betavoltaic batteries based on wide-bandgap semiconductor converters, a 4H-SiC p-n junction has been successfully utilized for fabricating a micro-nuclear battery with superior output performance due to high electron mobility, low dielectric constant, and robust radiation resistance.^{10–14} In 2015, Gui *et al.* conducted simulations on the performance of a 4H-SiC p-n junction betavoltaic battery using a ^{63}Ni source, revealing that increasing the thickness of the heavily p-type doped layer resulted in enhanced energy conversion efficiency.¹⁵ In 2016, Christopher Thomas *et al.* employed a Ti^3H_x film to develop a 4H-SiC p-n junction betavoltaic battery, employing an optimized 200 nm 4H-SiC p^+-n junction in the architecture.¹⁶ The implementation of this highly doped thin p^+-n junction design is considered as the most effective approach to improve the battery performance. In 2018, Kang *et al.* successfully fabricated an array consisting of 16 parallel-connected betavoltaic cells (BECs) with ^{63}Ni sources, which were composed of 4H-SiC p-i-n junctions and ^{63}Ni films.¹⁰ Experimental results demonstrated that when two BECs were connected in series or parallel, the corresponding open-circuit voltages were measured at 3.06 and 1.50 V, respectively. Furthermore, it was observed that the capacitor charging efficiency generated from two series-connected BECs reached $\sim 90.7\%$. Therefore, these findings indicated the main aspects in the structure of energy converters and the superior output performance of 4H-SiC-based betavoltaic batteries for promising applications in MEMS/NEMS. On the other hand, the previously reported results mainly focused on the 4H-SiC-based betavoltaic batteries with beta sources of low decay energy, such as ^3H and ^{63}Ni . So, utilizing high decay energy sources, such as the radioisotope ^{14}C , can potentially enable the development of 4H-SiC betavoltaic batteries and expand its practical applications.

In this work, the planar 4H-SiC p-n junction betavoltaic batteries with ^{14}C source films by using $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ were successfully developed. Through the utilization of the Monte Carlo method MCNPX code, we investigated the self-absorption effect of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source and the energy deposition distribution in the 4H-SiC energy converter. These investigations served as a basis for optimizing the source thickness and the irradiated electron-hole pair generation rate. Due to the intricate nature of the transportation of electron-hole pairs within the energy converter and their matching main energy deposition region, the COMSOL Multiphysics code was used to design realistic doping concentration of each layer along with its respective thickness, taking into account factors such as energy band structure, built-in potential barrier, depletion region width, minority carrier lifetime, and recombination rate. This approach allows for comprehensive consideration of internal factors affecting battery performance, including the energy conversion efficiency. Furthermore, a comprehensive model is developed to predict battery performance based on these results. Consequently, this research provides a feasible structure for a 4H-SiC p-n junction energy converter with a high decay energy ^{14}C source in the form of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$.

II. METHOD AND DEVICE STRUCTURE

A. Beta source

In general, the pure beta source ^{14}C is in the form of powder and can be transformed into diverse compound films.^{7–9} In this work, an ultra-thin film by using $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ was selected to design planar 4H-SiC p-n junction betavoltaic batteries. The $(\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ film has stable physical properties, such as being insoluble in water and insensitive to temperature. Furthermore, the chemical process in the specific labeling of ^{14}C atoms on the carbon atoms of two benzene rings within $(\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ is advanced and it can occur in continuous repetition, resulting in the formation of a planar ultra-thin film with a density of 1.21 g/cm^3 . Subsequently, the rectangular $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source is set to have a cross section of $1 \times 1\text{ cm}^2$. Its thickness increases with mass, and the original energy spectrum of the ^{14}C source is illustrated in Fig. 1. Then, utilizing the

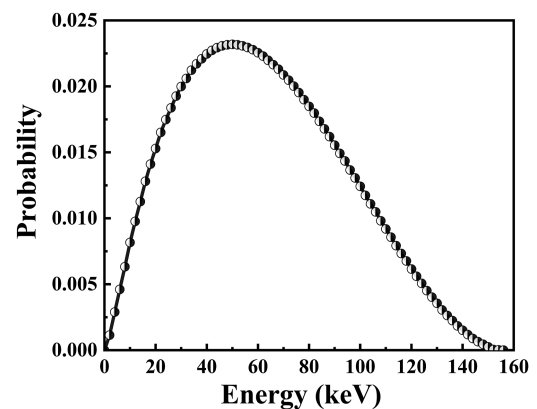


FIG. 1. Original energy spectrum of ^{14}C source.

Monte Carlo method MCNPX code, the self-absorption effect of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source is simulated and analyzed.

B. Energy deposition distribution

In a betavoltaic battery, the effective energy absorption region of a semiconductor energy converter is determined by the sum of the depletion region and the length of the minority carrier, which should be matched with the main energy deposition. By employing the Monte Carlo method MCNPX code, when the thickness of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source has a specific value, the accumulation deposited power density in the 4H-SiC energy converter can be obtained. Moreover, the irradiated electron-hole pair (EHP) generation rate $[G(x)]$ as a function of beta particle penetration depth in the 4H-SiC material decreases nearly exponentially as penetration depth increases and can be expressed as

$$G(x) = \frac{E_{dep}(x)}{E_{ehp}} = G_0 \exp(-\alpha x), \quad (1)$$

where $E_{dep}(x)$ is the energy deposition rate vs particle penetration depth, E_{ehp} is the average ionization energy of the semiconductor material and the experimental value is 7.8 eV for 4H-SiC,¹⁷ x denotes the particle penetration depth, and G_0 and α are the exponential parameters.

C. Energy converter structure

The structure of the 4H-SiC energy converter used in this work is depicted in Fig. 2(a), and it primarily consists of a p^+ -type 4H-SiC cap layer with a selected doping concentration of $1 \times 10^{19} \text{ cm}^{-3}$, a p^- -type 4H-SiC emission layer, an n^- -type 4H-SiC base layer, and an n^+ -type 4H-SiC buffer layer with a selected doping concentration of $1 \times 10^{16} \text{ cm}^{-3}$. In addition, the predominant regions for effective energy absorption are the p^- -type 4H-SiC emission layer and the n^- -type 4H-SiC base layer. Moreover, the high doping concentration of the p^+/n^+ -type 4H-SiC layers facilitates easy formation of stable ohmic contacts with metal conducting layers for upper and backside electrodes. Figure 2(b) illustrates the operational principle of a betavoltaic battery that serves as the fundamental basis for optimal structural design. The interaction between beta particles and semiconductor material generates numerous electron-hole pairs, which are subsequently separated by the built-in electric field before migrating toward the p-n junction boundary. Therefore, both the depletion region and the minority carrier length determine the effective energy absorption region.

D. Simulation and calculation method

For an energy converter, the charge collection efficiency of the irradiated electron-hole pairs in the depletion region can be considered as 100%, and the depletion region width is expressed as¹⁸

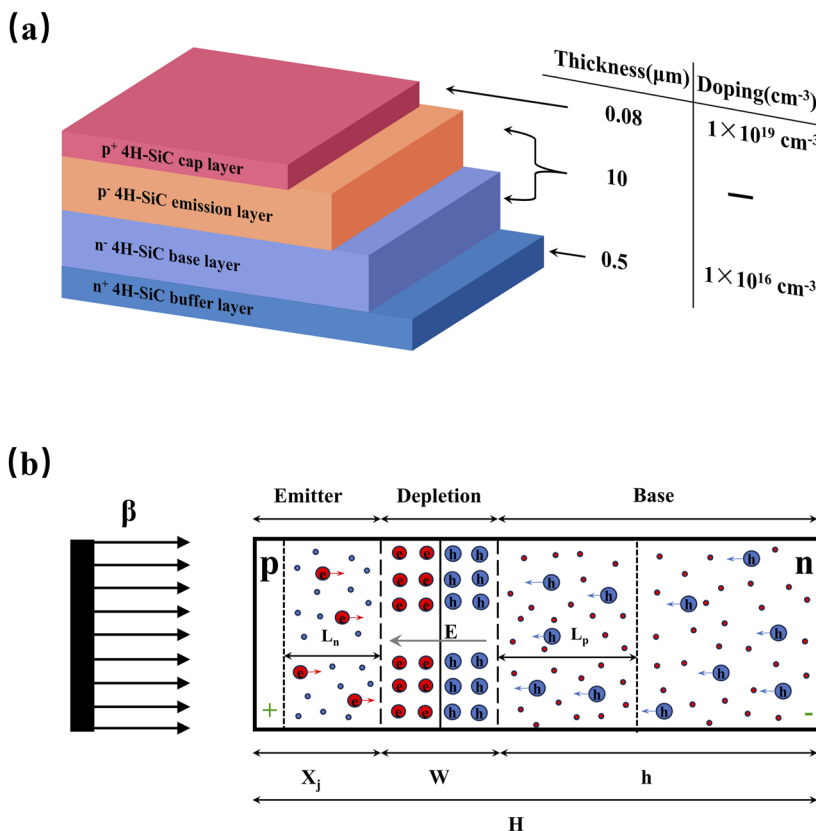


FIG. 2. (a) 3D structure of 4H-SiC energy converter. (b) Schematic diagram of 4H-SiC p-n junction betavoltaic battery.

$$W = \sqrt{2V_{bi} \frac{\epsilon_r \epsilon_0}{q} \frac{N_a + N_d}{N_a N_d}}, \quad (2)$$

where ϵ_0 ($\epsilon_r = 9.7$ for 4H-SiC)³ is the vacuum (relative) permittivity, q is the electron charge, and N_a (N_d) is the doping concentration in the p-type (n-type) region. V_{bi} is the built-in potential, which is considered as the theoretical limit of open-circuit voltage, and it is expressed as¹⁸

$$V_{bi} = \frac{KT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right), \quad (3)$$

where K is the Boltzmann constant, T is the absolute temperature, and n_i is the intrinsic carrier concentration. In addition, the minority carrier diffusion lengths in the n-type and p-type regions are described as L_n and L_p , respectively, which can be calculated using the following equations:¹⁹

$$L_n = \sqrt{\frac{KT}{q} \mu_n \tau_n} = \sqrt{\frac{KT}{q} \left[\frac{1020}{1 + \left(\frac{N_a + N_d}{1 + (1.5 \times 10^{17})} \right)^{0.61}} \left(\frac{T}{300} \right)^{-2} \right] \left[\frac{2.56 \times 10^{-6}}{1 + \left(\frac{N_a + N_d}{1 + (3 \times 10^{17})} \right)^{0.3}} \left(\frac{T}{300} \right)^{1.72} \right]}, \quad (4)$$

$$L_p = \sqrt{\frac{KT}{q} \mu_p \tau_p} = \sqrt{\frac{KT}{q} \left[\frac{112}{1 + \left(\frac{N_a + N_d}{1 + (4 \times 10^{18})} \right)^{0.4}} \left(\frac{T}{300} \right)^{-2} \right] \left[\frac{0.5 \times 10^{-6}}{1 + \left(\frac{N_a + N_d}{1 + (3 \times 10^{17})} \right)^{0.3}} \left(\frac{T}{300} \right)^{1.72} \right]}, \quad (5)$$

where τ_n (τ_p) is the electron (hole) lifetime and μ_n (μ_p) is the electron (hole) mobility. So, it is evident that the doping concentration in each layer plays a significant role in determining semiconductor performance.

Based on the transmission laws of beta particles emitted from a $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source in the 4H-SiC material, the COMSOL Multiphysics code is employed to model the behavior of irradiated non-equilibrium minority carriers. This employed model, which primarily incorporates the precipitation doping model, low-field mobility model, and Shockley–Read–Hall (SRH) composite model, determines the doping concentration and the thickness of each layer in the 4H-SiC p–n junction. Moreover, in the employed model, the coupled Poisson and carrier continuity equations are used to understand the physical transport and collection of irradiated non-equilibrium minority carriers, and they are expressed as follows:^{20–22}

$$\nabla^2 V = -\frac{\rho}{\epsilon_0 \epsilon_r}, \quad (6)$$

$$-\frac{1}{q} \nabla j_n = G(x) - R_n, \quad (7)$$

$$\frac{1}{q} \nabla j_p = G(x) - R_p, \quad (8)$$

where V is the electrostatic potential, ρ is the charge density, R_n (R_p) is the electron (hole) recombination rate, and j_n (j_p) is the electron (hole) current density. Furthermore, the electron (hole) current density is expressed as^{20–22}

$$j_n = -\frac{\mu_n KT}{q} \nabla n + \mu_n n \nabla V, \quad (9)$$

$$j_p = -\frac{\mu_p KT}{q} \nabla p + \mu_p p \nabla V, \quad (10)$$

where n (p) is the electron (hole) concentration.

The performance parameters of a betavoltaic battery, such as short-circuit current density (J_{sc}) and open-circuit voltage (V_{oc}), are determined by analyzing the J–V curve. Subsequently, the maximum output power density (P_m) can also be obtained. The fill factor (FF) denotes the interplay between the minority carrier collection and recombination processes, and it is expressed as¹⁸

$$FF = \frac{P_m}{J_{sc} V_{oc}}. \quad (11)$$

The energy conversion efficiency of semiconductor converter (η_c) is expressed as¹⁸

$$\eta_c = \frac{P_m}{P_{in}} = \frac{P_m}{AE_{ave}q}, \quad (12)$$

where A is the apparent source activity and E_{ave} denotes the average energy of the emitted β particles. Then, considering the self-absorption effect of the source, the overall battery efficiency (η) can be calculated.

III. RESULTS AND DISCUSSIONS

A. Calculations on beta source and energy deposition distribution

In Fig. 3(a), the emission power density and self-absorption rate as functions of source thickness for $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ are presented. It can be observed that, initially, there is a rapid increase in emission power density followed by a gradual ascent, ultimately reaching a saturation value close to $1.86 \mu\text{W}/\text{cm}^2$ at the thickness of $100 \mu\text{m}$.

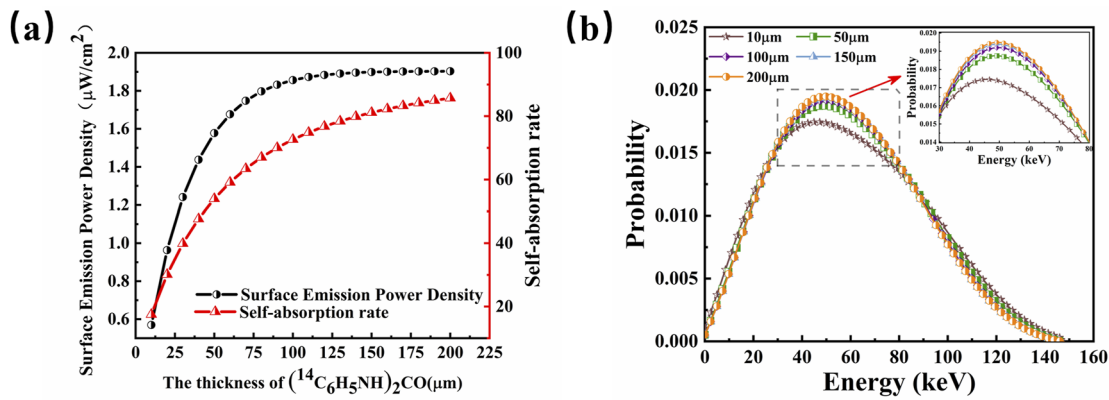


FIG. 3. (a) Emission power density and self-absorption rate vs thickness of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$. (b) Emission energy spectra of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ at specific thicknesses: 10, 50, 100, 150, and 200 μm .

In addition, with increasing thickness of the source material, its self-absorption rate also increases and reaches $\sim 72.7\%$ at the thickness of 100 μm . Figure 3(b) shows the emission energy spectra of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ at specific thicknesses: namely, 10, 50, 100, 150, and 200 μm . As the source thickness increases, the low-energy beta particles are more easily absorbed, resulting in only high energy beta particles escaping from the source. This leads to an overall shift in probability toward higher energy regions of the emitted energy spectrum. Finally, it should be noted that the decay energy of the source plays a crucial role in determining the radiation tolerance of converter materials and subsequently affects the life-span of betavoltaic batteries.

Figure 4 illustrates the accumulation deposited power density and the irradiated electron-hole pair generation rate as a function of beta particle penetration depth in the 4H-SiC material when the thickness of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ is selected at 100 μm . It can be calculated that the accumulation deposited power density exhibits a rapid

increase with increasing penetration depth until reaching a saturation value of $1.59 \mu\text{W}/\text{cm}^2$ at 40 μm in the 4H-SiC material. On the other hand, for the electron-hole pair (EHP) generation rate $G(x) = G_0 \exp(-\alpha x)$ as a function of beta particle penetration depth (x) in the 4H-SiC material, G_0 and α are calculated as $1.17 \times 10^{21} \text{ cm}^{-3} \text{ s}^{-1}$ and $1.03 \mu\text{m}^{-1}$, respectively.

B. Parameter optimization of energy converter

Considering the fabrication technology and preliminary scanning of structural parameters for a 4H-SiC p-n junction, the crucial regions for energy conversion are the p^- -type layer and the n^- -type layer. Aiming to achieve optimized doping concentration in both layers, the total thickness (H_{total}) of these layers remains constant at 10 μm in order to efficiently absorb the main energy from the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source. To maximize the maximum power density (P_m) of betavoltaic batteries, the specific thicknesses of the p^- -type layer (H_{p^-}) and n^- -type layer (H_{n^-}) are set at 0.8 and 9.2 μm , respectively. Based on the simulated results, the performance parameters of batteries vs the doping concentrations (N_a/N_d) in each layer are plotted in Fig. 5.

The analysis data indicate that the impact of doping concentration in the p^- -type region (N_a) on J_{sc} , P_m , η_c , and η is relatively insignificant compared with the doping concentration in the n^- -type region (N_d). However, when it comes to open-circuit voltage (V_{oc}), which has a theoretical limit value equal to the built-in potential (V_{bi}) of semiconductor converters, such as 4H-SiC p-n junctions, its value mainly depends on the doping concentration in each layer as shown in Fig. 5(b). Moreover, by optimizing the doping concentrations $N_a = 3 \times 10^{16} \text{ cm}^{-3}$ and $N_d = 2 \times 10^{14} \text{ cm}^{-3}$ within a range from $2 \times 10^{14} \text{ cm}^{-3}$ to $9 \times 10^{18} \text{ cm}^{-3}$, a maximum output power density (P_m) of $177.6 \text{ nW}/\text{cm}^2$ can be achieved. The other results correspond to the short-circuit current density (J_{sc}) of $104.9 \text{ nA}/\text{cm}^2$, the open-circuit voltage (V_{oc}) of 2.15 V, the fill factor (FF) of 78.8%, the energy conversion efficiency of semiconductor converter of 9.37%, and the overall battery efficiency of 2.56%. Therefore, both lightly doped N_a and heavily doped N_d contribute to enhancing battery performance.

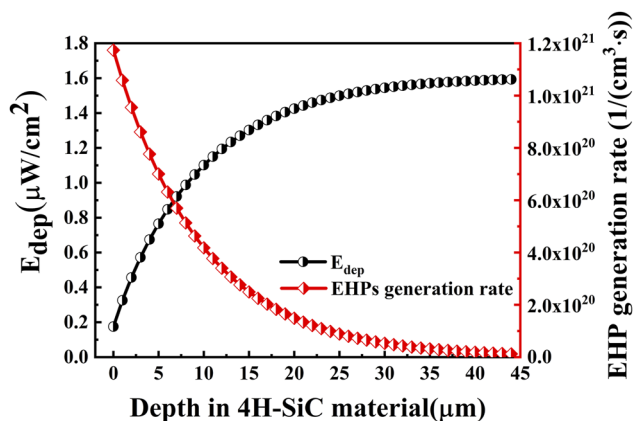


FIG. 4. Accumulation deposited power density and the electron-hole pair (EHP) generation rate vs penetration depth in 4H-SiC material for the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source at 100 μm .

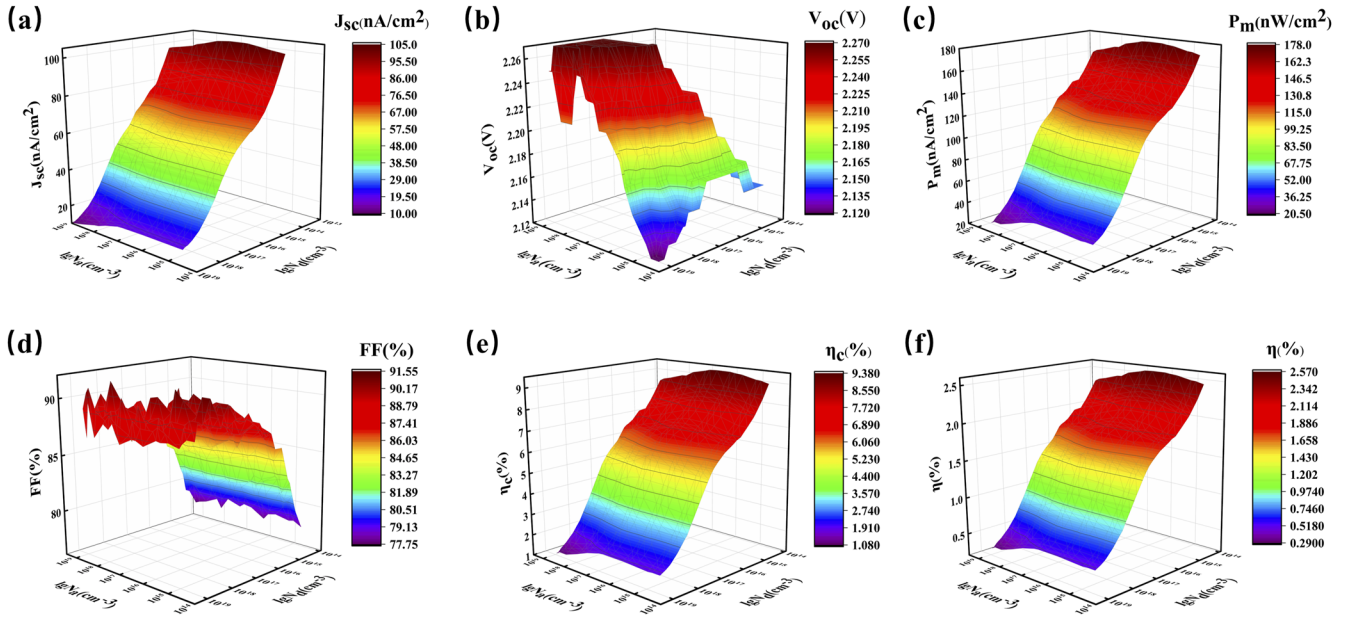


FIG. 5. Effects of N_a and N_d on (a) J_{sc} , (b) V_{oc} , (c) P_m , (d) FF, (e) η_c , (f) η . $H_{p-} = 0.8 \mu m$, and $H_{n-} = 9.2 \mu m$.

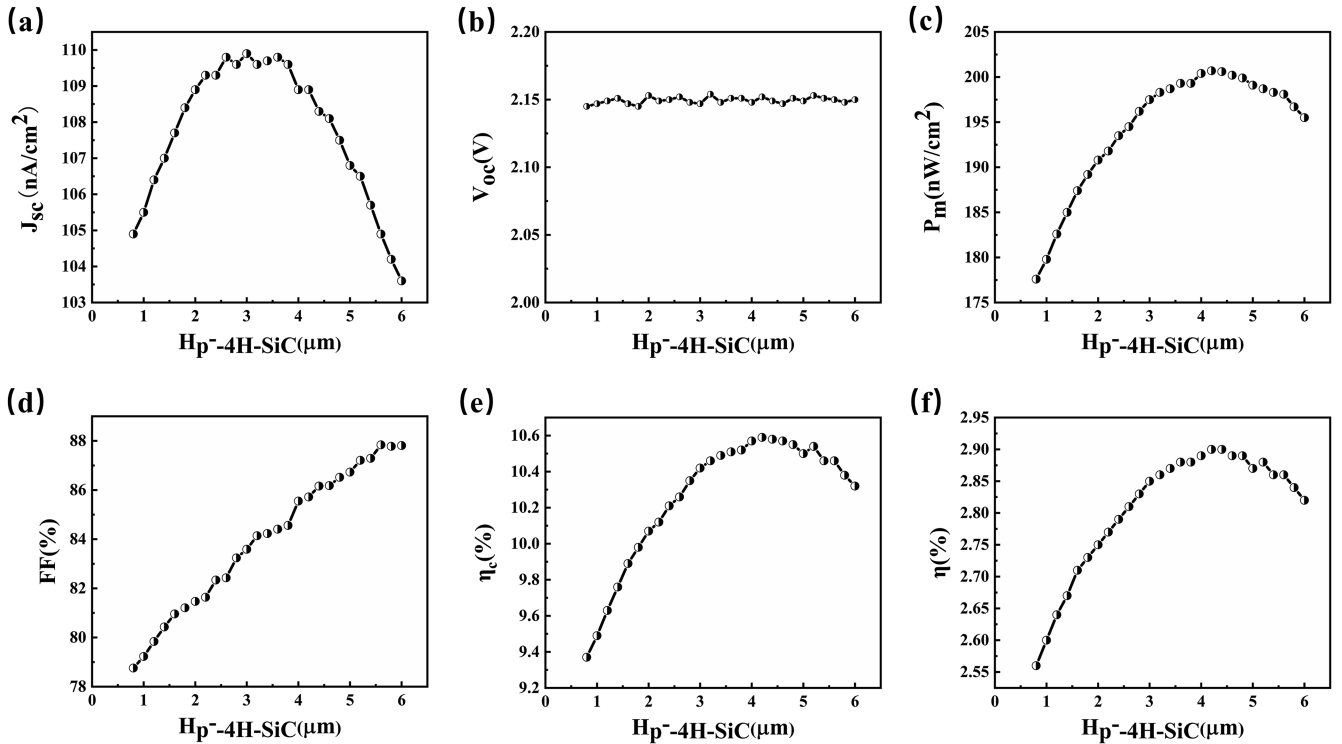


FIG. 6. Effects of H_{p-} on (a) J_{sc} , (b) V_{oc} , (c) P_m , (d) FF, (e) η_c , and (f) η . The doping concentrations $N_a = 3 \times 10^{16} \text{ cm}^{-3}$ in the p-type layer and $N_d = 2 \times 10^{14} \text{ cm}^{-3}$ in the n-type 4H-SiC base layer.

Then, the thickness of each layer should be designed. The high decay energy of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source results in a deep penetration depth. Therefore, by selecting an appropriate doping concentration for each layer, the width of the depletion region remains constant while adjusting the position of the depletion region from device surface. So, it is crucial to design a suitable thickness for each layer in the 4H-SiC p-n junction, particularly emphasizing on optimizing collection efficiency for induced electron-hole pairs in the p-type emission layer of the top junction.

As shown in Fig. 6, based on the initial structure with a total thickness $H = 10\text{ }\mu\text{m}$ and the doping concentrations $N_a = 3 \times 10^{16}\text{ cm}^{-3}$ in the p⁻-type layer and $N_d = 2 \times 10^{14}\text{ cm}^{-3}$ in the n⁻-type 4H-SiC base layer, the performance values of batteries are obtained by varying the thickness of the p⁻-type layer (H_{p-}) from 0.8 to 6 μm . It can be seen that V_{oc} remains almost unchanged as it is primarily determined by the doping concentration within the p-n junction. Notably, when H_{p-} reaches a value of 4.2 μm , the battery performance achieves the optimized values, including the short-circuit current density (J_{sc}) of 109 nA/cm², the fill factor (FF) of 85.7%, the energy conversion efficiency of semiconductor converter of 10.6%, and the overall battery efficiency of 2.9%. Furthermore, it should be mentioned that for the low-energy beta sources, such as ^{63}Ni and ^3H , used in a 4H-SiC betavoltaic battery, the thinner emission layer generally results in improved performance due to their limited penetration depth within a few micrometers for the collection of induced electron-hole pairs in/near the depletion region. However, considering that the energy deposition depth of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source extends to 44 μm in the 4H-SiC material, both the depletion region and the diffusion length of the minority carrier region play significant roles in achieving collection efficiency for induced electron-hole pairs. Finally, the optimized structural parameters of a 4H-SiC p-n junction and the corresponding performance values are obtained for the ^{14}C ultra-thin film source by using $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$.

IV. CONCLUSION

In summary, this study utilizes the Monte Carlo code and the COMSOL Multiphysics code to design planar 4H-SiC p-n junctions for the high-energy pure beta source ^{14}C in the form of $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ films. In addition, a comprehensive model is constructed to predict the performance of batteries. The saturation thickness of the $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source reaches nearly 100 μm due to its high decay energy, resulting in the maximum emitted power density and the deep penetration depth of beta particles in the 4H-SiC material. The systematic investigation demonstrated that incorporating a lightly doped thick p⁻-type layer in the top junction is a feasible approach to mitigate carrier recombination beyond the depletion region for enhanced performance. This conclusion differs from the conventional semiconductor converter structure with low-energy sources. Finally, an optimized 4H-SiC -($^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ betavoltaic battery is proposed considering real semiconductor process. For this proposal, selecting a 100 μm -thick $(^{14}\text{C}_6\text{H}_5\text{NH})_2\text{CO}$ source, an optimized structure consists of a 4.2 μm -thick p⁻-type region with a doping concentration of $N_a = 3 \times 10^{16}\text{ cm}^{-3}$ and a 5.8 μm -thick n⁻-type region with a doping concentration of $N_d = 2 \times 10^{14}\text{ cm}^{-3}$ for the 4H-SiC p-n junction as an energy converter. This configuration achieves a maximum power density of

0.2 $\mu\text{W}/\text{cm}^2$ and an energy conversion efficiency of semiconductor converter of 10.6%. Furthermore, realizing a high-energy betavoltaic battery using a 4H-SiC p-n junction holds promise for developing high-performance micro-nuclear batteries based on ^{14}C sources characterized by long half-life and high output power.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Weikai Yuan: Conceptualization (equal); Data curation (equal); Investigation (equal); Software (equal); Writing – original draft (equal). **Guodong Wei:** Conceptualization (equal); Investigation (equal); Writing – review & editing (equal). **Yumin Liu:** Conceptualization (equal); Methodology (equal); Writing – original draft (equal). **Houjun He:** Conceptualization (equal); Investigation (equal). **Xiaoyan Li:** Conceptualization (equal); Investigation (equal). **Guanghui Zhang:** Conceptualization (equal); Methodology (equal). **Zhuoyang Lv:** Conceptualization (equal); Investigation (equal). **Jiahui Zhang:** Conceptualization (equal); Investigation (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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